The given problem is about the planta física challenge at Tecnológico de Monterrey, where we have to divide the monthly demand between the potable water and the reusable water along the 4 buildings that are (AULAS 1, BIBLIOTECA,CEDETEC, AULAS 2),

We define a chromosome as a vector of four integer levels, $k = (k_1, k_2, k_3, k_4)$. Each position in the chromosome corresponds strictly to a building (k_1 = AULAS 1, k_2 = BIBLIOTECA, k_3 = CEDETEC, k_4 = AULAS 2).

To translate the genetic representation into physical quantities, we distinguish three layers of the decision:

The integer level(x_i) is the value chosen by the algorithm within the allowable domain for each building

Fraction(x_i) each level is 10 percentage points for the reuse fraction calculated as $x_i = \frac{k_i}{10}$

Volume(T_i & P_i) for the water(m^3) For the treated water in each building $T_i = D_i(\frac{k_i}{10})$ and the remaining potable water volume is $P_i = D_i - T_i$

Now we define the total demand for the campus as Q = 2276.21 in March 2025 (data given in the base kit) and the capacity the campus can satisfy is C = 1410 m^3

The breakdown per building, domains and maximum reuse limits are summarized below:

Building	Demand D (m^3)	Domain of k
Aulas 1	847.00	{0,...,8}
Biblioteca	462.00	{0,...,8}
Cedetec	284.21	{0,...,6}
Aulas 2	683.00	{0,...,8}

And the formulas for calculating the optimal solutions are

- $T = \sum_{i=1}^{4}$, Total treated water
- $P = Q - T$, Total potable water
- $E = max\{0, T - C\}$ Excess capacity

Objective and fitness directions

$$\begin{cases} \frac{P}{Q}, T \leq C \\ 1 + \frac{E}{C}, T > C \end{cases}$$

Since total campus demand exceeds the treatment plant's capacity, we seek to optimize the distribution across all buildings. Lower fitness values represent better candidates because they correspond to a lower proportion of potable water consumed (when feasible). Minimizing potable water usage is crucial, as relying on external sources beyond plant capacity increases operational costs and external dependency.

Calculate one candidate by hand.

k	x	D	T	P
6	0.6	847	508.20	338.8
6	0.6	462	277.2	184.8
6	0.6	284.21	170.526	113.684
6	0.6	683	409.8	273.2

- Total treated Water(T)

$$\sum_{i=1}^{4} T = 508.20 + 277.20 + 170.526 + 409.8$$

$$T = 1365.726$$

- Total Potable Water(P)

$$\sum_{i=1}^{4} P = 338.80 + 184.80 + 113.684 + 273.2 = 910.484$$

- Total campus demand T + P = Q

$$1365.726 + 910.484 = 2276.21$$

- Excess Capacity(E) C = 1410m^3

$$E = max(0, T - c) = max\{0, 1365.726 - 1410\}$$

$$E = max\{0, -44.274\} = 0m^3$$

- Feasibility Decision

$$T \leq C \ if \ feasible, \ (1365.726 \leq 1410)$$

$$F = \frac{P}{Q} = \frac{910.484}{2276.21} = 0.4$$

In accordance with the research instructions, we execute the base cells containing NumPy imports, data storage, and the evaluate function. We call the function using our baseline candidate vector k = (6, 6, 6, 6).

Treated	Potable	Excess	Feasible	Fitness
0	2276.21	0.0	True	1
1365.725999	910.4840000000002	0.0	True	0.4
1764.1260000000002	910.4840000000002	354.1260000000002	False	1.2511531914893619

Comparing our manual calculations against the five keys returned by the evaluate function (treated, potable, excess, feasible), we construct the following comparison table:

Metric Key	Manual Calculation Value	Evaluator Output Value
Treated Water (T)	$1365.726m^3$	$1365.726m^3$
Potable Water (P)	$910.484\ m^3$	$910.4840000000002\ m^3$
Excess Capacity (E)	0	0
Feasibility	True	True
Fitness(F)	0.4	0.4000000000000001

The complete numerical agreement between our manual calculations and the evaluate function output validates the correctness of the provided base environment. Minor floating-point variations are strictly attributable to standard IEEE 754 precision representation in NumPy and have no operational impact on feasibility decisions. This verification confirms that the implementation faithfully executes the intended mathematical model.

Analysis of the boundary case $k = (8, 8, 6, 8)$ highlights the critical distinction between individual gene validity and system-wide infrastructure limits. While every building respects its permitted domain, the aggregate demand of $1764.126m^3$ exceeds the campus treatment plant capacity, triggering the penalization branch ($F = 1 + E/C$). This robust penalty structure successfully guides the optimization process away from infrastructural bottlenecks, setting a reliable foundation for the upcoming Development phase.